

G.A.N.E NAV — Comprehensive User Requirements & Bug Report

Current Issues Reported

1. Buttons & UI Elements Not Working

- ☐ Buttons are empty — no text/content visible
- ☐ Buttons don't respond to clicks
- ☐ Missing button labels and descriptions
- ☐ Buttons appear but are non-functional

2. Text & Content Issues

- ☐ Empty spaces where text should be
- ☐ Missing labels on interactive elements
- ☐ Inconsistent text across windows/panels
- ☐ No descriptions for features

3. Color & Visual Consistency

- ☐ Colors don't match across different windows/panels
- ☐ Visual inconsistency between UI sections
- ☐ Theme not applied uniformly

4. Notification System Issues

- ☐ Warnings loop repeatedly (annoying)
- ☐ Warnings appear every second — too frequent
- ☐ No option to mute/silence notifications
- ☐ No way to disable unnecessary alerts
- ☐ Need: Option to disable repetitive warnings

5. Interface Complexity

- ☐ Overly complex interface

- ☐ Not intuitive for users
- ☐ Too many options without clear organization

6. Category Selection & Options

- ☐ No category selection options
- ☐ No choices available in categories
- ☐ Missing dropdowns/selectors for categories
- ☐ Can't select vehicle type to match route

7. Vehicle Type Selection

- ☐ **CRITICAL:** No option to select vehicle type
- ☐ Can't specify vehicle dimensions (length, width, height)
- ☐ Can't set vehicle weight class (light, medium, heavy)
- ☐ Routes not optimized for vehicle type
- ☐ No vehicle-specific restrictions (height, weight, cargo)
- ☐ Need: Vehicle profile system with:
 - Vehicle type (car, truck, bus, motorcycle, etc.)
 - Dimensions (length, width, height)
 - Weight (empty, loaded)
 - Cargo type
 - Axle count
 - Special restrictions

8. Route Optimization for Vehicles

- ☐ Routes don't account for vehicle size
- ☐ Routes don't account for vehicle weight
- ☐ Routes don't account for vehicle restrictions
- ☐ No vehicle-specific routing logic
- ☐ Heavy trucks can't avoid low bridges/tunnels
- ☐ Oversized vehicles can't avoid narrow roads

- ☐ No height/width/weight restrictions in routing

9. Missing Features from Previous Requests

- ☐ APK generation (waiting for production URL)
- ☐ Offline map regions (added 8 regions, need testing)
- ☐ PWA install prompt (added, needs testing)
- ☐ Responsive design for mobile/tablet (added, needs testing)
- ☐ DrawingToolbar integration (fixed, needs testing)

Historical Requirements (From All Conversations)

Project Vision

- **Purpose:** Advanced navigation system for Israel/Palestine region with quantum interface design
- **Target Users:** Professional drivers, fleet managers, emergency services, military/security operations
- **Design Philosophy:** Futuristic, high-visibility, quantum aesthetic with vivid colors and depth

Core Navigation Features

- Real-time GPS navigation with multiple routing modes (drive, walk, emergency, plan)
- Offline maps with tile caching and region pre-download
- Multi-modal routing (car, public transport, walking, cycling)
- Traffic visualization and congestion detection
- Route history and favorites
- Voice commands and audio guidance
- Gesture-based controls

Map & Visualization

- Multiple map layers (OSM, satellite, terrain, topographic, dark mode, hybrid)
- Weather radar overlay with real-time data
- Satellite imagery with drawing tools

- Indoor positioning system
- AR navigation
- 3D terrain visualization
- V2X (Vehicle-to-Everything) communication layer
- Digital twin visualization
- GNSS constellation management
- Traffic camera feeds

Advanced Features

- AI route optimization with machine learning
- Driver scoring system
- Fleet management dashboard
- Real-time collaboration (multiple users on same map)
- Offline mode with delta updates
- Payment system integration (Stripe)
- Evidence/incident reporting
- Smart alerts and risk assessment
- Analytics engine
- Command center for operations
- EOC (Emergency Operations Center) panel
- Accessibility features (A11y)

Data & Integration

- Real-time data streams from multiple sources
- Integration with external APIs (weather, traffic, maps)
- WebSocket for real-time updates
- OAuth authentication
- Database for user profiles, routes, preferences
- S3 storage for media/documents

Cross-Platform Requirements

- **Browser:** Works on all modern browsers (Chrome, Safari, Firefox, Edge)
- **Mobile:** Responsive design for phones (portrait/landscape)
- **Tablet:** Optimized layout for tablets
- **Desktop:** Full-featured desktop experience
- **PWA:** Installable as app on home screen
- **APK:** Native Android app via Bubblewrap/TWA
- **Offline:** Full functionality without internet
- **Auto-adaptation:** Automatic layout adjustment to screen size, resolution, orientation

Language & Localization

- **Hebrew:** Primary language (all UI elements in Hebrew)
- **English:** Secondary language
- **Arabic:** Support for Arabic-speaking users
- **Bilingual:** Seamless switching between languages
- **RTL Support:** Right-to-left text for Hebrew/Arabic

Performance & Reliability

- Fast response times (< 100ms for UI interactions)
- Smooth animations and transitions
- No lag or stuttering
- Reliable offline functionality
- Automatic reconnection when network returns
- Data persistence across sessions
- No data loss

Security & Privacy

- User authentication with OAuth
- Encrypted data transmission
- Secure offline storage
- No tracking without consent
- GDPR compliance

- Data minimization

Accessibility

- Keyboard navigation
- Screen reader support
- High contrast mode
- Reduced motion support
- Large text support
- Touch-friendly targets (min 48x48px)

Design Quality

- Professional, polished UI
- Consistent design language
- High-quality graphics and icons
- Smooth animations
- Responsive to all screen sizes
- Dark/light theme support
- Accessible color contrasts

Priority Fixes (Current Session)

P0 — Critical (Blocking Usage)

1. **Fix empty buttons** — ensure all buttons have visible text and are clickable
2. **Fix missing labels** — add text to all UI elements
3. **Add vehicle type selection** — implement vehicle profile system
4. **Fix color consistency** — ensure colors match across all panels
5. **Fix notification spam** — reduce frequency, add mute option

P1 — High (Major Features)

1. **Vehicle-specific routing** — optimize routes based on vehicle type/dimensions/weight
2. **Category selection** — add dropdowns/selectors for all categories
3. **Fix responsive design** — ensure mobile/tablet layout works correctly

4. **Test offline maps** — verify 16 regions download and cache correctly
5. **Test PWA install** — verify install prompt appears and works

P2 — Medium (Enhancements)

1. **Generate APK** — create Android app with Bubblewrap
2. **Add real-time traffic layer** — integrate Mapbox/HERE traffic
3. **Improve panel organization** — simplify interface, reduce complexity
4. **Add feature descriptions** — help text for all features
5. **Performance optimization** — reduce load times, smooth animations

Technical Debt & Fixes Applied

-  Fixed 16 TypeScript errors (EOCPanel, FuturisticFeatures, SpecVaultPanel)
-  Fixed Vite WebSocket HMR error (proxy environment)
-  Fixed DrawingToolbar sidebar integration
-  Added PWA manifest.json with icons
-  Added responsive CSS system
-  Added useDeviceAdaptation hook
-  Added 8 offline map regions
-  Fixed WeatherRadarOverlay retry logic
-  All 438 tests passing
-  APK generation (waiting for production URL)

Next Steps

1. Fix all P0 issues immediately
2. Implement vehicle type selection system
3. Test all features on real devices
4. Generate APK and publish to Google Play
5. Collect user feedback and iterate